

Your Name (first last):

Names of peers you worked with:

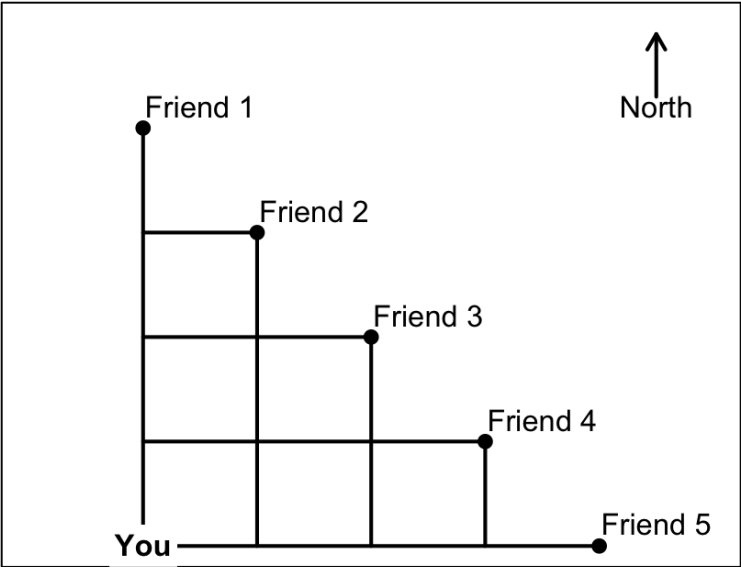


Figure 1. The map of where you live relative to your friends. If you flip a coin and get heads, you go north; if you get tails, you go east.

1. Complete the table of outcomes below to calculate all possible combinations of heads and tails for 4 coin flips. The table has been partially completed to demonstrate the process.

Start	First flip	Second flip	Third flip	Fourth flip	Ordered outcome	Number "heads"	Friend visited
You	H	H	H	H	HHHH	4	1
				T	HHHT	3	
			T				
		T					
	T						

2. Next, flip your own coin four times, then record the outcome on the board. Which friend did you visit?

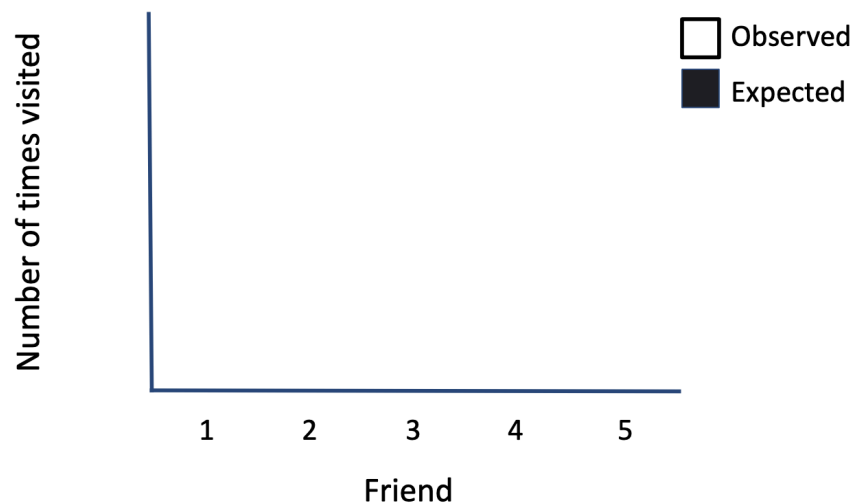
3. Complete the table below. You need to calculate the probability of each event (# of visits to each friend out of total visits - see table on page 1), determine the expected number of individuals based on the probability and the number of students who participated in the activity (I will tell you the value of n once everyone is in class), record the actual number of friends visited (Observed column) based on the coin flipping data from all students, and calculate the squared difference between the observed and expected values divided by expected.

Friend	Probability	Expected for n =	Observed (from class)	$\frac{(Obs_i - Exp_i)^2}{Exp_i}$
1				
2				
3				
4				
5				

Sum of last column (= test statistic called the chi-square) =

P value (see table on lecture slide) =

4. Make a paired barplot of the observed and expected data. Add numbers to the y axis.



5. Last, write a claim based on the results (chi-square value, p value, barplot). Did the observed data differ statistically from the expectations?